Verificación de Programas Concurrentes Usando Lógica

Logical Foundations of Concurrent Computation

Jorge A. Pérez
University of Groningen, The Netherlands
j.a.perez [[at]] rug.nl

ECI 2026
(Version of July 27, 2026)

Preambulo

Outline

Verificación de Programas Concurrentes

- Muchos sistemas informáticos son **concurrentes**, y están compuestos de programas que se ejecutan simultáneamente.

Verificación de Programas Concurrentes

- Muchos sistemas informáticos son **concurrentes**, y están compuestos de programas que se ejecutan simultáneamente.
- **Ejemplos:**
  - Sistemas distribuidos y paralelos
  - Arquitecturas basadas en microservicios
  - Go / Erlang / Rust / Akka / Elixir

Verificación de Programas Concurrentes

- Muchos sistemas informáticos son **concurrentes**, y están compuestos de programas que se ejecutan simultáneamente.
- **Ejemplos:**
  - Sistemas distribuidos y paralelos
  - Arquitecturas basadas en microservicios
  - Go / Erlang / Rust / Akka / Elixir
- Mecanismo de coordinación entre programas: **paso de mensajes**.

Verificación de Programas Concurrentes

- Muchos sistemas informáticos son **concurrentes**, y están compuestos de programas que se ejecutan simultáneamente.
- **Ejemplos:**
  - Sistemas distribuidos y paralelos
  - Arquitecturas basadas en microservicios
  - Go / Erlang / Rust / Akka / Elixir
- Mecanismo de coordinación entre programas: **paso de mensajes**.
- Establecer la correctitud de estos programas implica garantizar que las interacciones entre ellos respetan los **protocolos establecidos**.

Verificación de Programas Concurrentes

- Muchos sistemas informáticos son **concurrentes**, y están compuestos de programas que se ejecutan simultáneamente.
- **Ejemplos:**
  - Sistemas distribuidos y paralelos
  - Arquitecturas basadas en microservicios
  - Go / Erlang / Rust / Akka / Elixir
- Mecanismo de coordinación entre programas: **paso de mensajes**.
- Establecer la correctitud de estos programas implica garantizar que las interacciones entre ellos respetan los **protocolos establecidos**.
- Una tarea crucial, pero también compleja: necesitamos técnicas de verificación capaces de analizar **estructuras de comunicación** sofisticadas.

Verificación de Programas Concurrentes **Usando Lógica**

Este Curso

- Técnicas fundamentales para la verificación para programas concurrentes.
- **Enfasis:** Conceptos de la **lógica** para el análisis y verificación de programas.

Verificación de Programas Concurrentes **Usando Lógica**

Este Curso

- Técnicas fundamentales para la verificación para programas concurrentes.
- **Enfasis:** Conceptos de la **lógica** para el análisis y verificación de programas.

Ideas Principales

- Describir los protocolos de comunicación usando **sistemas de tipos**.
- Especificar los programas concurrentes con un **lenguaje formal de procesos**.
- Los tipos aseguran que los procesos **usan sus recursos correctamente**.

$\rightarrow$ Uso adecuado de recursos = ejecución correcta de protocolos.

Verificación de Programas Concurrentes **Usando Lógica**

Este Curso

- Técnicas fundamentales para la verificación para programas concurrentes.
- **Enfasis:** Conceptos de la **lógica** para el análisis y verificación de programas.

Ideas Principales

- Describir los protocolos de comunicación usando **sistemas de tipos**.
- Especificar los programas concurrentes con un **lenguaje formal de procesos**.
- Los tipos aseguran que los procesos **usan sus recursos correctamente**.

$\rightarrow$ Uso adecuado de recursos = ejecución correcta de protocolos.

Correspondencia de Curry-Howard

Demostrar un teorema y escribir un programa correcto son la **misma actividad** desde perspectivas distintas.

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Ejemplo: Un Programa con Paso de Mensajes

Una función con dos parámetros (dos canales)

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Recibe un mensaje en `c1`, continúa en `d1`

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Envía '41' en c2, continúa en d2

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Espera confirmación para cerrar canal `d1`

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Envía mensaje para cerrar canal d2

Ejemplo: Un Programa con Paso de Mensajes

```
peter c1 c2 =
  let (x, d1) = receive c1 in
  let d2 = send (41) c2 in
  wait d1;
  close d2;
  ()
```

Retorna valor de tipo `unit`

Ejemplo: Dos Programas con Paso de Mensajes

```
peter c1 c2 =
  let (x, c1) = receive c1 in
  let c2 = send (41) c2 in
  wait c1;
  close c2;
  ()
```

```
miles c1 c2 =
  let (x, c2) = receive c2 in
  let c1 = send (42) c1 in
  wait c2;
  close c1;
  ()
```

Ejemplo: Dos Programas con Paso de Mensajes

```
peter c1 c2 =
  let (x, c1) = receive c1 in
  let c2 = send (41) c2 in
  wait c1;
  close c2;
  ()
```

```
miles c1 c2 =
  let (x, c2) = receive c2 in
  let c1 = send (42) c1 in
  wait c2;
  close c1;
  ()
```

El programa respeta su protocolo ✔

Ejemplo: Dos Programas con Paso de Mensajes

```
peter c1 c2 =
  let (x, c1) = receive c1 in
  let c2 = send (41) c2 in
  wait c1;
  close c2;
  ()
```

El programa respeta su protocolo ✔

```
miles c1 c2 =
  let (x, c2) = receive c2 in
  let c1 = send (42) c1 in
  wait c2;
  close c1;
  ()
```

El programa también respeta su protocolo ✔

Ejemplo: Interacción de Programas

```
peter c1 c2 =
  let (x, c1) = receive c1 in
  let c2 = send (41) c2 in
  wait c1;
  close c2;
  ()
```

```
miles c1 c2 =
  let (x, c2) = receive c2 in
  let c1 = send (42) c1 in
  wait c2;
  close c1;
  ()
```

```
main =
  let (c1, c1dual) = new @T () in
  let (c2, c2dual) = new @T () in
  fork (miles c1dual c2);
       peter c1 c2dual
```

Es ‘`main`’ correcto?

😐

Verificación de Programas usando Sistemas de Tipos

Programas (Rust, Go, ...)

Procesos

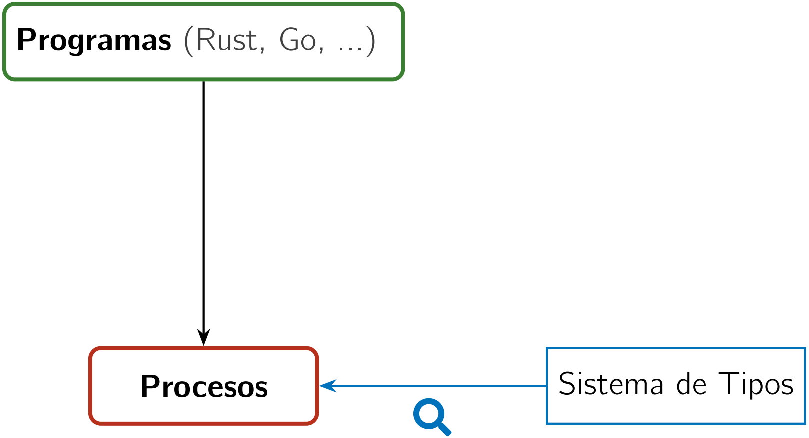
Verificación de Programas usando Sistemas de Tipos

Programas (Rust, Go, ...)

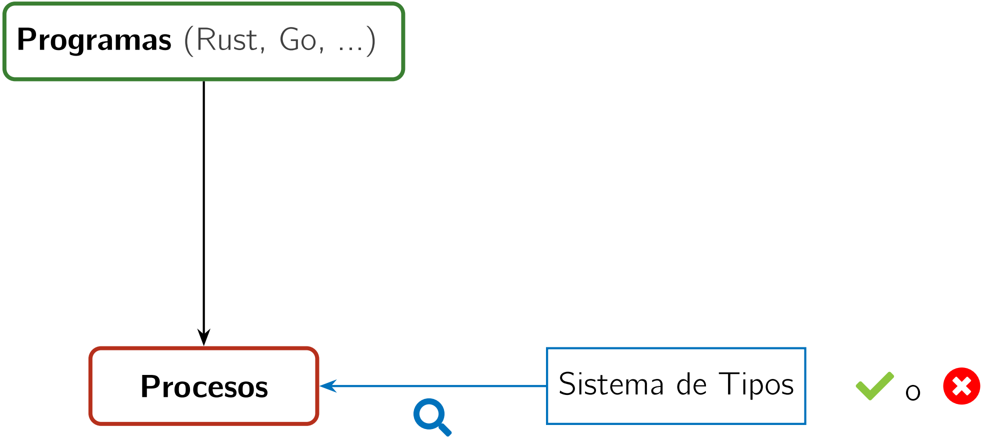
$\downarrow$

Procesos

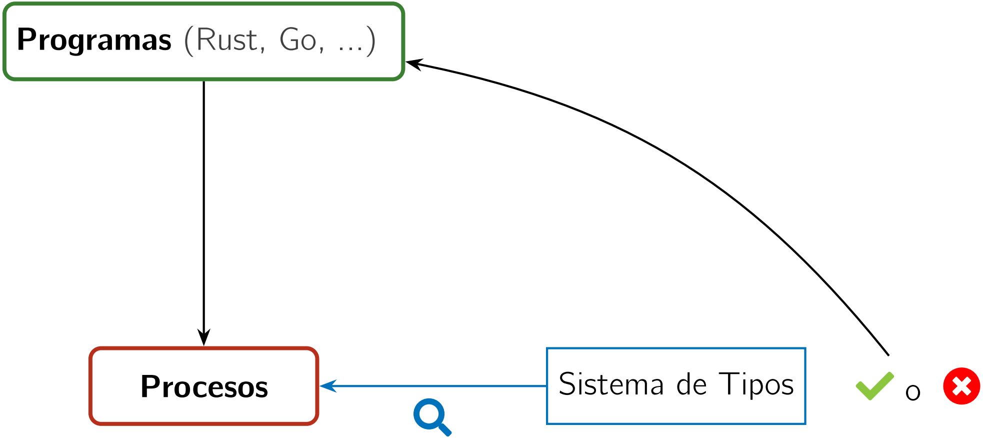
Verificación de Programas usando Sistemas de Tipos



Verificación de Programas usando Sistemas de Tipos



Verificación de Programas usando Sistemas de Tipos



Outline

Un Lenguaje Formal para Concurrencia: Cálculos de Procesos

Asumiendo un conjunto de **nombres** $x, y, z, \ldots$ (usados para representar **canales**), definimos un lenguaje formal de **procesos** (denotados $P, Q, \ldots$):

Un Lenguaje Formal para Concurrencia: Cálculos de Procesos

Asumiendo un conjunto de **nombres** $x, y, z, \ldots$ (usados para representar **canales**), definimos un lenguaje formal de **procesos** (denotados $P, Q, \ldots$):

$$
\begin{array}{rlll}
P, Q ::= & \overline{x}\langle y\rangle.P & \textbf{envia} \text{ mensaje } y \text{ en el canal } x\text{, continúa con } P \\
\mid & x(y).P & \textbf{recibe} \text{ un mensaje } z \text{ en el canal } x\text{, continúa con } P\{^{z}/_{y}\}
\end{array}
$$

Un Lenguaje Formal para Concurrencia: Cálculos de Procesos

Asumiendo un conjunto de **nombres** $x, y, z, \ldots$ (usados para representar **canales**), definimos un lenguaje formal de **procesos** (denotados $P, Q, \ldots$):

$P, Q ::=$	$\overline{x}\langle y\rangle.P$	**envia** mensaje y en el canal x, continúa con P
$\mid$	$x(y).P$	**recibe** un mensaje z en el canal x, continúa con $P\{^z/_y\}$
$\mid$	$\overline{x}\langle\rangle.P$	**envía** una señal para cerrar el canal x, continúa como P
$\mid$	$x().P$	**recibe** señal para cerrar el canal x, continúa como P

Un Lenguaje Formal para Concurrencia: Cálculos de Procesos

Asumiendo un conjunto de **nombres** $x, y, z, \ldots$ (usados para representar **canales**), definimos un lenguaje formal de **procesos** (denotados P, $Q, \ldots$):

$$
\begin{array}{rlll}
P, Q ::= & \overline{x}\langle y\rangle.P & & \textbf{envia} \text{ mensaje } y \text{ en el canal } x\text{, continúa con } P \\
| & x(y).P & & \textbf{recibe} \text{ un mensaje } z \text{ en el canal } x\text{, continúa con } P\{^z/_y\} \\
| & \overline{x}\langle\rangle.P & & \textbf{envía} \text{ una señal para cerrar el canal } x\text{, continúa como } P \\
| & x().P & & \textbf{recibe} \text{ señal para cerrar el canal } x\text{, continúa como } P \\
| & P \mid Q & & \textbf{concurrencia}\text{: } P \text{ y } Q \text{ se ejecutan en paralelo} \\
| & (\boldsymbol{\nu} x)P & & \textbf{restricción}\text{: el canal } x \text{ es local (privado) en } P \\
| & 0 & & \textbf{cero}\text{: el proceso que no hace nada}
\end{array}
$$

Un Lenguaje Formal: Cálculos de Procesos

Sintaxis de procesos:

$$P, Q ::= \overline{x}\langle y\rangle.P \;\Big|\; x(y).P \;\Big|\; \overline{x}\langle\rangle.P \;\Big|\; x().P \;\Big|\; P \mid Q \;\Big|\; (\boldsymbol{\nu} x)P \;\Big|\; 0$$

Cómo expresar interacción?

Un Lenguaje Formal: Cálculos de Procesos

Sintaxis de procesos:

$$P, Q ::= \overline{x}\langle y\rangle.P \;\Big|\; x(y).P \;\Big|\; \overline{x}\langle\rangle.P \;\Big|\; x().P \;\Big|\; P \mid Q \;\Big|\; (\boldsymbol{\nu} x)P \;\Big|\; 0$$

Cómo expresar interacción?

La **relación de reducción** formaliza la comunicación entre procesos concurrentes.

Dos reglas de la definición:

$$\overline{x}\langle z\rangle.P \mid x(y).Q \longrightarrow P \mid Q\{^{z}/_{y}\}$$

$$\overline{x}\langle\rangle.P \mid x().Q \longrightarrow P \mid Q$$

Un Lenguaje Formal: Cálculos de Procesos

Sintaxis de procesos:

$$P, Q ::= \overline{x}\langle y\rangle.P \;\Big|\; x(y).P \;\Big|\; \overline{x}\langle\rangle.P \;\Big|\; x().P \;\Big|\; P \mid Q \;\Big|\; (\boldsymbol{\nu} x)P \;\Big|\; 0$$

Cómo expresar interacción?

La **relación de reducción** formaliza la comunicación entre procesos concurrentes.

Dos reglas de la definición:

$$\overline{x}\langle z\rangle.P \mid x(y).Q \longrightarrow P \mid Q\{^{z}/_{y}\}$$

$$\overline{x}\langle\rangle.P \mid x().Q \longrightarrow P \mid Q$$

Intuición:

Acciones que ocurren en paralelo (y en el mismo canal) pueden dar lugar a una sincronización. Las acciones deben ser complementarias (input / output).

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$

3) $\overline{x}\langle\rangle.x().P$

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$

no hay reducción:

las acciones no son complementarias

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$

3) $\overline{x}\langle\rangle.x().P$

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$

no hay reducción:

las acciones no son complementarias

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$

no hay reducción:

los canales son diferentes

3) $\overline{x}\langle\rangle.x().P$

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$

no hay reducción:

las acciones no son complementarias

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$

no hay reducción:

los canales son diferentes

3) $\overline{x}\langle\rangle.x().P$

no hay reducción:

las acciones son secuenciales

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$ — no hay reducción: las acciones no son complementarias

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$ — no hay reducción: los canales son diferentes

3) $\overline{x}\langle\rangle.x().P$ — no hay reducción: las acciones son secuenciales

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$ — hay dos reducciones

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

1) $x(z).P \mid x(u).Q$ — no hay reducción: las acciones no son complementarias

2) $\overline{x}\langle z\rangle.P \mid z(u).Q$ — no hay reducción: los canales son diferentes

3) $\overline{x}\langle\rangle.x().P$ — no hay reducción: las acciones son secuenciales

4) $\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q$ — hay dos reducciones

5) $\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0$ — hay una reducción: el canal z es un mensaje!

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

Veamos (4) en detalle:

$$\begin{aligned}\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q &\longrightarrow x(z).P \mid \overline{x}\langle v_2\rangle.Q\{{}^{v_1}/_{y}\}\\ &\longrightarrow P\{{}^{v_2}/_{z}\} \mid Q\{{}^{v_1}/_{y}\}\end{aligned}$$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

Veamos (4) en detalle:

$$\begin{aligned}\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q &\longrightarrow x(z).P \mid \overline{x}\langle v_2\rangle.Q\{^{v_1}/_y\}\\ &\longrightarrow P\{^{v_2}/_z\} \mid Q\{^{v_1}/_y\}\end{aligned}$$

(5) es interesante: hay dos posibles reducciones, pero sólo una puede ocurrir. Por un lado tenemos:

$$\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0 \longrightarrow 0 \mid z(u).Q_1\{^{z}/_y\} \mid x(y).y().0$$

Ejercicio: Cuáles procesos pueden reducir (interactuar)?

Veamos (4) en detalle:

$$\begin{aligned}\overline{x}\langle v_1\rangle.x(z).P \mid x(y).\overline{x}\langle v_2\rangle.Q &\longrightarrow x(z).P \mid \overline{x}\langle v_2\rangle.Q\{^{v_1}/_y\}\\ &\longrightarrow P\{^{v_2}/_z\} \mid Q\{^{v_1}/_y\}\end{aligned}$$

(5) es interesante: hay dos posibles reducciones, pero sólo una puede ocurrir. Por un lado tenemos:

$$\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0 \longrightarrow 0 \mid z(u).Q_1\{^{z}/_y\} \mid x(y).y().0$$

pero también:

$$\overline{x}\langle z\rangle.0 \mid x(y).y(u).Q_1 \mid x(y).y().0 \longrightarrow 0 \mid x(y).y(u).Q_1 \mid z().0$$

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0$$

$$|$$

$$d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)$$

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0$$

Proceso de `peter`

$$|$$

$$d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)$$

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0$$

Proceso de `peter`

$$|$$

$$d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)$$

Proceso de `miles`

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0$$

Proceso de `peter`

$$|$$

$$d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)$$

Proceso de `miles`

El análisis del sistema completo nos permite detectar una dependencia circular entre los dos sub-procesos: el programa no puede reducir y llega a un **deadlock**.

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$
\begin{array}{c}
(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0 \\
\mid \\
d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)
\end{array}
$$

Proceso de `peter`

Proceso de `miles`

El análisis del sistema completo nos permite detectar una dependencia circular entre los dos sub-procesos: el programa no puede reducir y llega a un **deadlock**.

Un Lenguaje Formal: Cálculos de Procesos

Regresando al programa `main`, el protocolo de comunicación correspondiente se puede expresar de forma formal y compacta usando procesos:

$$(\boldsymbol{\nu} c)(\boldsymbol{\nu} d)(c(x).\overline{d}\langle 41\rangle.c().\overline{d}\langle\rangle.0$$

Proceso de `peter`

$$|$$

$$d(x).\overline{c}\langle 42\rangle.d().\overline{c}\langle\rangle.0)$$

Proceso de `miles`

El análisis del sistema completo nos permite detectar una dependencia circular entre los dos sub-procesos: el programa no puede reducir y llega a un **deadlock**.

Este curso

Presentamos el uso de lógica para la verificación estática de procesos concurrentes, asegurando que los protocolos se ejecutan correctamente, sin llegar a *deadlocks*.

Outline

Contexto y Motivación

Un Lenguaje Formal para Concurrencia

Plan del Curso

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.
2. **Linear Logic (LL)**: Una lógica para el uso correcto de recursos computacionales.

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.
2. **Linear Logic (LL)**: Una lógica para el uso correcto de recursos computacionales.
3. La **correspondencia de Curry-Howard** para procesos concurrentes:

$$\begin{array}{rcl} \text{proposiciones en LL} & \Longleftrightarrow & \text{session types} \\ \text{pruebas en LL} & \Longleftrightarrow & \text{procesos concurrentes} \\ \text{simplificación de pruebas} & \Longleftrightarrow & \text{sincronización entre procesos} \end{array}$$

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.
2. **Linear Logic (LL)**: Una lógica para el uso correcto de recursos computacionales.
3. La **correspondencia de Curry-Howard** para procesos concurrentes:

$$\begin{array}{rcl} \text{proposiciones en LL} & \Longleftrightarrow & \text{session types} \\ \text{pruebas en LL} & \Longleftrightarrow & \text{procesos concurrentes} \\ \text{simplificación de pruebas} & \Longleftrightarrow & \text{sincronización entre procesos} \end{array}$$

4. La correspondencia usando LL en su **versión clásica**.

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.
2. **Linear Logic (LL)**: Una lógica para el uso correcto de recursos computacionales.
3. La **correspondencia de Curry-Howard** para procesos concurrentes:

$$\begin{array}{rcl} \text{proposiciones en LL} & \Longleftrightarrow & \text{session types} \\ \text{pruebas en LL} & \Longleftrightarrow & \text{procesos concurrentes} \\ \text{simplificación de pruebas} & \Longleftrightarrow & \text{sincronización entre procesos} \end{array}$$

4. La correspondencia usando LL en su **versión clásica**.
5. Comunicación asíncrona y deadlock freedom.

Plan del Curso

1. **Session types**: Especificaciones precisas de protocolos para canales.
2. **Linear Logic (LL)**: Una lógica para el uso correcto de recursos computacionales.
3. La **correspondencia de Curry-Howard** para procesos concurrentes:

$$\begin{array}{rcl} \text{proposiciones en LL} & \Longleftrightarrow & \text{session types} \\ \text{pruebas en LL} & \Longleftrightarrow & \text{procesos concurrentes} \\ \text{simplificación de pruebas} & \Longleftrightarrow & \text{sincronización entre procesos} \end{array}$$

4. La correspondencia usando LL en su **versión clásica**.
5. Comunicación asíncrona y deadlock freedom.

---

6. Extensión de la correspondencia con recursos ilimitados: **clientes y servidores**.
7. La lógica de **implicaciones agrupadas** (*bunched implications*).

Aspectos Prácticos

Cada clase estará dividida en tres partes:

- 1:30 - 2:15: Parte 1
- 2:15 - 2:20: Pausa
- 2:20 - 3:00: Parte 2
- 3:00 - 3:20: Coffee break
- 3:20 - 4:30: Parte 3

Material asociado al curso, actualizado después de cada clase:

`https://jperez.nl/teaching/eci26`

Preguntas, sugerencias, consultas:

`j.a.perez@rug.nl` (subject: [eci26])

Introduction

Linear Logic

Outline

Propositions as Types: The Standard Story

Linear Logic for Resource Management

Sequent Calculus for ILL

From Intuitionistic to Linear Logic

Modeling Examples

Intuitionistic Logic (IL)

The grammar of propositions:

$$A, B \quad ::= \quad X \mid A \rightarrow B \mid A \times B \mid A + B$$

where:

- X is a propositional **constant**
- $A \rightarrow B$ denotes **implication**
- $A \times B$ denotes **conjunction** (alternative notation for $A \land B$)
- $A + B$ denotes **disjunction** (alternative notation for $A \lor B$)

Intuitionistic Logic (IL)

The grammar of propositions:

$$A, B \quad ::= \; X \mid A \rightarrow B \mid A \times B \mid A + B$$

where:

- X is a propositional **constant**
- $A \rightarrow B$ denotes **implication**
- $A \times B$ denotes **conjunction** (alternative notation for $A \wedge B$)
- $A + B$ denotes **disjunction** (alternative notation for $A \vee B$)

Assumptions and judgments:

- Γ, Δ denote **assumptions**: sequences of zero or more propositions.
- A **judgment** $\Gamma \vdash A$ means "from assumptions Γ we can conclude A".

Rules for Intuitionistic Logic (IL)

Proving judgments of the form $\Gamma \vdash A$.

- A single **axiom**, a rule without premises.

Rules for Intuitionistic Logic (IL)

Proving judgments of the form $\Gamma \vdash A$.

- A single **axiom**, a rule without premises.
- **Structural rules** control the ordering, duplication, and discarding of hypotheses in the assumption Γ.

Rules for Intuitionistic Logic (IL)

Proving judgments of the form $\Gamma \vdash A$.

- A single **axiom**, a rule without premises.
- **Structural rules** control the ordering, duplication, and discarding of hypotheses in the assumption Γ.
- Each logical connective is specified in terms of **two rules**:
  an introduction rule and an elimination rule (this is called **natural deduction**).

Rules for Intuitionistic Logic (IL)

Proving judgments of the form $\Gamma \vdash A$.

- A single **axiom**, a rule without premises.
- **Structural rules** control the ordering, duplication, and discarding of hypotheses in the assumption Γ.
- Each logical connective is specified in terms of **two rules**:
  an introduction rule and an elimination rule (this is called **natural deduction**).
- Using the rules, we aim to construct **derivations** for judgements.

Rules for Intuitionistic Logic (IL)

Proving judgments of the form $\Gamma \vdash A$.

- A single **axiom**, a rule without premises.
- **Structural rules** control the ordering, duplication, and discarding of hypotheses in the assumption Γ.
- Each logical connective is specified in terms of **two rules**:
  an introduction rule and an elimination rule (this is called **natural deduction**).
- Using the rules, we aim to construct **derivations** for judgements.

(Later we will see rules in a **sequent calculus**, rather than in natural deduction.)

Rules for IL

Id

$$\frac{}{A \vdash A}$$

Exchange

$$\frac{\Gamma, \Delta \vdash A}{\Delta, \Gamma \vdash A}$$

Contraction

$$\frac{\Gamma, A, A \vdash B}{\Gamma, A \vdash B}$$

Weakening

$$\frac{\Gamma \vdash B}{\Gamma, A \vdash B}$$

$\to$-I

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}$$

$\to$-E

$$\frac{\Gamma \vdash A \to B \qquad \Delta \vdash A}{\Gamma, \Delta \vdash B}$$

$\times$-I

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \times B}$$

$\times$-E

$$\frac{\Gamma \vdash A \times B \qquad \Delta, A, B \vdash C}{\Gamma, \Delta \vdash C}$$

$+$-I$_1$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A + B}$$

$+$-I$_2$

$$\frac{\Gamma \vdash B}{\Gamma \vdash A + B}$$

$+$-E

$$\frac{\Gamma \vdash A + B \qquad \Delta, A \vdash C \qquad \Delta, B \vdash C}{\Gamma, \Delta \vdash C}$$

Example of a Derivation

A derivation for the judgment

$$A \to B, A \vdash A \times B$$

expressed as a **proof tree** (labels for the rules are omitted):

$$\dfrac{\dfrac{\dfrac{\dfrac{}{A \vdash A} \qquad \dfrac{\dfrac{}{A \to B \vdash A \to B} \qquad \dfrac{}{A \vdash A}}{A \to B, A \vdash B}}{A, A \to B, A \vdash A \times B}}{A \to B, A, A \vdash A \times B}}{A \to B, A \vdash A \times B}$$

We see how A is used twice, with the help of contraction and exchange rules.

Alternative Rules

We have just seen:

$$\times\text{-I}\quad \frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \times B} \qquad\qquad \times\text{-E}\quad \frac{\Gamma \vdash A \times B \qquad \Delta, A, B \vdash C}{\Gamma, \Delta \vdash C}$$

We can also have:

$$\times\text{-I'}\quad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \times B} \qquad \times\text{-E}_1\quad \frac{\Gamma \vdash A \times B}{\Gamma \vdash A} \qquad \times\text{-E}_2\quad \frac{\Gamma \vdash A \times B}{\Gamma \vdash B}$$

The rules are **equivalent**: they are interderivable using structural rules.

More Alternative Rules

We have just seen:

$$\text{Id}\ \frac{}{A \vdash A} \qquad \to\text{-I}\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B} \qquad \to\text{-E}\ \frac{\Gamma \vdash A \to B \qquad \Delta \vdash A}{\Gamma, \Delta \vdash B}$$

We can also have:

$$\text{Id'}\ \frac{}{\Gamma, A, \Delta \vdash A} \qquad \to\text{-I}\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B} \qquad \to\text{-E'}\ \frac{\Gamma \vdash A \to B \qquad \Gamma \vdash A}{\Gamma \vdash B}$$

Here again, structural rules ensure that either group of rules can be used.
We prefer to have an explicit treatment of structural rules.

Commuting Conversions

The order in which certain (structural) rules are applied is irrelevant.

An equivalence relation on proofs, denoted $\Longleftrightarrow$, via **commuting conversions**.

Here are the commuting conversions involving contraction and $\to$-E:

$$\frac{}{\Gamma, C, \Delta \vdash B} \quad \Longleftrightarrow \quad \frac{}{\Gamma, C, \Delta \vdash B}$$

$$\frac{}{\Gamma, \Delta, C \vdash B} \quad \Longleftrightarrow \quad \frac{}{\Gamma, \Delta, C \vdash B}$$

Commuting Conversions

The order in which certain (structural) rules are applied is irrelevant.

An equivalence relation on proofs, denoted $\Longleftrightarrow$, via **commuting conversions**.

Here are the commuting conversions involving contraction and $\rightarrow$-E:

$$\dfrac{\dfrac{\Gamma, C, C \vdash A \rightarrow B}{\Gamma, C \vdash A \rightarrow B} \qquad \Delta \vdash A}{\Gamma, C, \Delta \vdash B} \quad \Longleftrightarrow \quad \dfrac{\dfrac{\Gamma, C, C \vdash A \rightarrow B \qquad \Delta \vdash A}{\Gamma, C, C, \Delta \vdash B}}{\Gamma, C, \Delta \vdash B}$$

$$\dfrac{}{\Gamma, \Delta, C \vdash B} \quad \Longleftrightarrow \quad \dfrac{}{\Gamma, \Delta, C \vdash B}$$

Commuting Conversions

The order in which certain (structural) rules are applied is irrelevant.

An equivalence relation on proofs, denoted $\Longleftrightarrow$, via **commuting conversions**.

Here are the commuting conversions involving contraction and $\to$-E:

$$\dfrac{\dfrac{\Gamma, C, C \vdash A \to B}{\Gamma, C \vdash A \to B} \qquad \Delta \vdash A}{\Gamma, C, \Delta \vdash B} \quad \Longleftrightarrow \quad \dfrac{\dfrac{\Gamma, C, C \vdash A \to B \qquad \Delta \vdash A}{\Gamma, C, C, \Delta \vdash B}}{\Gamma, C, \Delta \vdash B}$$

$$\dfrac{\Gamma \vdash A \to B \qquad \dfrac{\Delta, C, C \vdash A}{\Delta, C \vdash A}}{\Gamma, \Delta, C \vdash B} \quad \Longleftrightarrow \quad \dfrac{\dfrac{\Gamma \vdash A \to B \qquad \Delta, C, C \vdash A}{\Gamma, \Delta, C, C \vdash B}}{\Gamma, \Delta, C \vdash B}$$

Proof Reduction

- A judgement may have several, very different derivations.

Proof Reduction

- A judgement may have several, very different derivations.
- In particular, it is not very useful to have an introduction rule for a connective that is followed by the corresponding elimination rule.

Proof Reduction

- A judgement may have several, very different derivations.
- In particular, it is not very useful to have an introduction rule for a connective that is followed by the corresponding elimination rule.
- The interest is in **reducing** proofs to avoid unnecessary occurrence of connectives; the proofs themselves may not be "smaller".

Proof Reduction

- A judgement may have several, very different derivations.
- In particular, it is not very useful to have an introduction rule for a connective that is followed by the corresponding elimination rule.
- The interest is in **reducing** proofs to avoid unnecessary occurrence of connectives; the proofs themselves may not be "smaller".
- Notions of **normal form** and **subformula property** apply for proofs and their reductions.

Proof Reduction: Example

Consider the proof

$$\dfrac{\dfrac{\dfrac{\dfrac{\overline{B \vdash B} \qquad \overline{B \vdash B}}{B, B \vdash B \times B}}{B \vdash B \times B}\,(\dagger)}{\vdash B \to (B \times B)} \qquad \dfrac{\overline{A \to B \vdash A \to B} \qquad \overline{A \vdash A}}{A \to B, A \vdash B}}{A \to B, A \vdash B \times B}$$

It can be reduced to:

$$\dfrac{\dfrac{\dfrac{\overline{A \to B \vdash A \to B} \qquad \overline{A \vdash A}}{A \to B, A \vdash B} \qquad \dfrac{\overline{A \to B \vdash A \to B} \qquad \overline{A \vdash A}}{A \to B, A \vdash B}}{A \to B, A, A \to B, A \vdash B \times B}}{A \to B, A \vdash B \times B}\,(\dagger)$$

Propositions as Types

Extending judgements to contain terms (= programs!).

- From the logic perspective, terms encode proofs.
- From the programmer's perspective, terms are a programming language for which logic provides a **type system**.

Propositions as Types

Propositions can be read as types, and proofs are read as corresponding typed terms:

- A proof of $A \to B$ is a **function** that takes any proof of A into a proof of B.
- A proof of $A \times B$ is a **pair** of a proof of A and a proof of B.
- A proof of $A + B$ is a **disjoint sum**: either a proof of A or a proof of B.

Terms encode the rules of the logic:

- An introduction rule: a term that **constructs** a value.
- An elimination rule: a term that **deconstructs** a value.
- There are no term forms for the structural rules.

We briefly have a look at the terms (λ-calculus!) and their reductions.

Intuitionistic Terms

$$
\begin{aligned}
s, t, u, v, w \quad ::= \quad & x \\
& \mid \lambda x.\, u \mid s\,(t) \\
& \mid (t, u) \mid \mathsf{case}\ s\ \mathsf{of}\ (x, y) \to v \\
& \mid \mathtt{inl}\,(t) \mid \mathtt{inr}\,(u) \mid \mathsf{case}\ s\ \mathsf{of}\ \mathtt{inl}\,(x) \to v;\ \mathtt{inr}\,(y) \to w
\end{aligned}
$$

Assumptions Γ, Δ are now written in the form

$$x_1 : A_1, \ldots, x_n : A_n$$

where $n \geq 0$ and

- $x_1, \ldots, x_n$ are distinct variables;
- $A_1, \ldots, A_n$ are types (= propositions).

Judgments are then of the form $\Gamma \vdash t : A$ ("term t has type A").

Types for Intuitionistic Terms

Id
$$\frac{}{x : A \vdash x : A}$$

Exchange
$$\frac{\Gamma, \Delta \vdash A}{\Delta, \Gamma \vdash A}$$

Contraction
$$\frac{\Gamma, y : A, z : A \vdash u : B}{\Gamma, x : A \vdash u[x/y, x/z] : B}$$

Weakening
$$\frac{\Gamma \vdash u : B}{\Gamma, x : A \vdash u : B}$$

$\to$-I
$$\frac{\Gamma, x : A \vdash u : B}{\Gamma \vdash \lambda x.\, u : A \to B}$$

$\to$-E
$$\frac{\Gamma \vdash s : A \to B \qquad \Delta \vdash t : A}{\Gamma, \Delta \vdash s\,(t) : B}$$

$\times$-I
$$\frac{\Gamma \vdash t : A \qquad \Delta \vdash u : B}{\Gamma, \Delta \vdash (t, u) : A \times B}$$

$\times$-E
$$\frac{\Gamma \vdash s : A \times B \qquad \Delta, x : A, y : B \vdash v : C}{\Gamma, \Delta \vdash \mathsf{case}\ s\ \mathsf{of}\ (x, y) \to v : C}$$

Types for Intuitionistic Terms

+-I$_1$

$$\frac{\Gamma \vdash t : A}{\Gamma \vdash \mathtt{inl}\,(t) : A + B}$$

+-I$_2$

$$\frac{\Gamma \vdash u : B}{\Gamma \vdash \mathtt{inr}\,(u) : A + B}$$

+-E

$$\frac{\Gamma \vdash s : A + B \qquad \Delta, x : A \vdash v : C \qquad \Delta, y : B \vdash w : C}{\Gamma, \Delta \vdash \mathsf{case}\; s\; \mathsf{of}\; \mathtt{inl}\,(x) \to v;\; \mathtt{inr}\,(y) \to w : C}$$

Typing Derivations: Example

Recall the derivation for the judgment $A \to B, A \vdash A \times B$, seen before. Now with terms:

$$
\dfrac{\dfrac{\dfrac{}{y : A \vdash y : A} \qquad \dfrac{\dfrac{}{f : A \to B \vdash f : A \to B} \qquad \dfrac{}{z : A \vdash z : A}}{f : A \to B, z : A \vdash f\,(z) : B}}{y : A, f : A \to B, z : A \vdash (y, f\,(z)) : A \times B}}{\dfrac{f : A \to B, y : A, z : A \vdash (y, f\,(z)) : A \times B}{f : A \to B, x : A \vdash (x, f\,(x)) : A \times B}}
$$

Term Reduction

Since terms encode proofs, proof reductions in logic correspond to term reductions:

$$
\begin{aligned}
\lambda x.\, u\,(t) &\Longleftrightarrow u[t/x] \\
\mathtt{case}\,(t, u)\,\mathtt{of}\,(x, y) \to v &\Longleftrightarrow u[t/x, v/y] \\
\mathtt{case}\,\mathtt{inl}\,(t)\,\mathtt{of}\ \mathtt{inl}\,(x) \to v;\ \mathtt{inr}\,(y) \to w &\Longleftrightarrow v[t/x] \\
\mathtt{case}\,\mathtt{inr}\,(u)\,\mathtt{of}\ \mathtt{inl}\,(x) \to v;\ \mathtt{inr}\,(y) \to w &\Longleftrightarrow w[u/x]
\end{aligned}
$$

Notice:
The notion of proof reduction, associated to properties of proof systems, corresponds precisely to term reduction for a model of computation!

Outline

Propositions as Types: The Standard Story

Linear Logic for Resource Management

Sequent Calculus for ILL

From Intuitionistic to Linear Logic

Modeling Examples

Linear Logic: A Substructural Logic for Computation

- Introduced by Jean-Yves Girard in 1987.
- **Observation**: Contraction arbitrarily duplicates assumptions; weakening freely discards them. Not always sensible for computational resources!

Linear Logic: A Substructural Logic for Computation

- Introduced by Jean-Yves Girard in 1987.
- **Observation**: Contraction arbitrarily duplicates assumptions; weakening freely discards them. Not always sensible for computational resources!
- Linear logic advocates a **disciplined** use of resources, by controlling contraction and weakening. Many applications in proof theory and computer science.
- A "resource-aware" logic:
  - $A \vdash B$: given A-resources, a way to obtain B-resources
  - Proof theoretically: **substructural logic**

Linear Logic

- We are mostly interested in the intuitionistic variant, dubbed ILL.
  We will also have a look at classical LL (CLL).
- We see rules as a sequent calculus, rather than as natural deduction.
  - In natural deduction: introduction and elimination rules
  - In sequent calculi: left and right rules, plus the cut rule

Linear Logic: Propositions

'A is true' vs 'I have A-resources'

$A \to B$	if A is true, then B is true
$A \wedge B$	both A and B are true

Linear Logic: Propositions

'A is true' vs 'I have A-resources'

$A \rightarrow B$	if A is true, then B is true
$A \wedge B$	both A and B are true

$A \multimap B$	if you give me A, then I can produce B

Linear Logic: Propositions

'A is true' vs 'I have A-resources'

$A \rightarrow B$	if A is true, then B is true
$A \wedge B$	both A and B are true
$A \multimap B$	if you give me A, then I can produce B
$A \otimes B$	I have both A and B

Linear Logic: Propositions

'A is true' vs 'I have A-resources'

$A \rightarrow B$	if A is true, then B is true
$A \wedge B$	both A and B are true

$A \multimap B$	if you give me A, then I can produce B
$A \otimes B$	I have both A and B
$A \mathbin{\&} B$	Choose from A and B

Examples:

- $\mathit{euro} \otimes \mathit{euro} \multimap \mathit{pizza}$
- $\mathit{dough} \otimes (\mathit{sauce} \otimes \mathit{toppings}) \multimap \mathit{pizza}$

Linear Logic: Linearity

$$A \to A \land A \qquad A \not\multimap A \otimes A$$

$$A \land B \to A \qquad A \otimes B \not\multimap A$$

Linear Logic: Linearity

$$A \to A \wedge A \qquad\qquad A \not\multimap A \otimes A$$

$$A \wedge B \to A \qquad\qquad A \otimes B \not\multimap A$$

$$A \wedge (A \to B) \to B \qquad\qquad A \otimes (A \multimap B) \multimap B$$

Linear Logic: Linearity

$$A \to A \land A \qquad A \not\multimap A \otimes A$$

$$A \land B \to A \qquad A \otimes B \not\multimap A$$

$$A \land (A \to B) \to B \qquad A \otimes (A \multimap B) \multimap B$$

$$A \land (A \to B) \to A \land B \qquad A \otimes (A \multimap B) \not\multimap A \otimes B$$

Linear Logic is **substructural**, resource-aware.

Outline

Proof Theory of $\wedge$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A}$$

$$\wedge\mathsf{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B}$$

$$\wedge\mathsf{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

$$\mathsf{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C}$$

$$\mathsf{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}$$

$$\mathsf{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\wedge$

Assertion A holds under the list of hypotheses Γ.
Intuitively, $\bigwedge \Gamma \implies A$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \wedge R\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B} \qquad \wedge L\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

$$\text{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C} \qquad \text{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C} \qquad \text{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\wedge$

Axiom rule

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A}$$

$\wedge$R

$$\frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B}$$

$\wedge$L

$$\frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

Weakn

$$\frac{\Gamma \vdash C}{\Gamma, A \vdash C}$$

Contr

$$\frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}$$

eXchg

$$\frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\wedge$

Right and left rules for $\wedge$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A}$$

$\wedge$R
$$\frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B}$$

$\wedge$L
$$\frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

Weakn
$$\frac{\Gamma \vdash C}{\Gamma, A \vdash C}$$

Contr
$$\frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}$$

eXchg
$$\frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\wedge$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \wedge\mathrm{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B} \qquad \wedge\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

$$\underbrace{\text{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C} \qquad \text{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C} \qquad \text{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}}_{\boxed{\text{Structural rules}}}$$

Proof Theory of $\wedge$

$\Gamma : \mathit{Frml} \to \mathbb{N}$ is a multiset

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \wedge\mathsf{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B} \qquad \wedge\mathsf{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

$$\mathsf{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C} \qquad \mathsf{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C} \qquad \mathsf{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

We can omit the exchange rule by treating Γ as a multiset

Generalized Structural Rules

$$\frac{\Gamma, \Gamma \vdash C}{\Gamma \vdash C} \qquad \frac{\Gamma \vdash C}{\Gamma, \Delta \vdash C}$$

Obtained from repeatedly using the usual rules (and exchange).

Using Structural Rules

$$\frac{}{A \wedge B \vdash A} \qquad \frac{}{A \vdash A \wedge A}$$

Using Structural Rules

$$\dfrac{\overline{A, B \vdash A}}{A \land B \vdash A}$$

$$\dfrac{}{A \vdash A \land A}$$

Using Structural Rules

$$\dfrac{\dfrac{A \vdash A}{A, B \vdash A}}{A \wedge B \vdash A}$$

$$\dfrac{}{A \vdash A \wedge A}$$

Using Structural Rules

$$\dfrac{\dfrac{A \vdash A}{A, B \vdash A}}{A \wedge B \vdash A}$$

$$\dfrac{\dfrac{}{A, A \vdash A \wedge A}}{A \vdash A \wedge A}$$

Using Structural Rules

$$\dfrac{\dfrac{A \vdash A}{A, B \vdash A}}{A \wedge B \vdash A} \qquad\qquad \dfrac{\dfrac{A \vdash A \qquad A \vdash A}{A, A \vdash A \wedge A}}{A \vdash A \wedge A}$$

Using Structural Rules

$$\dfrac{\dfrac{A \vdash A}{A, B \vdash A}}{A \wedge B \vdash A} \qquad \dfrac{\dfrac{A \vdash A \qquad A \vdash A}{A, A \vdash A \wedge A}}{A \vdash A \wedge A}$$

The usage of the contraction and weakening rules is crucial!

Associativity

$$\frac{}{A \wedge (B \wedge C) \vdash (A \wedge B) \wedge C}$$

Associativity

$$\dfrac{\dfrac{}{A, B, C \vdash (A \wedge B) \wedge C}}{A \wedge (B \wedge C) \vdash (A \wedge B) \wedge C}$$

Associativity

$$\dfrac{\dfrac{\dfrac{}{A, B \vdash A \wedge B} \qquad C \vdash C}{A, B, C \vdash (A \wedge B) \wedge C}}{A \wedge (B \wedge C) \vdash (A \wedge B) \wedge C}$$

Associativity

$$\dfrac{\dfrac{\dfrac{A \vdash A \qquad B \vdash B}{A, B \vdash A \wedge B} \qquad C \vdash C}{A, B, C \vdash (A \wedge B) \wedge C}}{A \wedge (B \wedge C) \vdash (A \wedge B) \wedge C}$$

Alternative Rules for $\wedge$

We can replace $\wedge L$ with the following two rules:

$$\wedge\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C} \quad \longrightarrow \quad \wedge\mathrm{L}_1\ \frac{\Gamma, A \vdash C}{\Gamma, A \wedge B \vdash C} \qquad \wedge\mathrm{L}_2\ \frac{\Gamma, B \vdash C}{\Gamma, A \wedge B \vdash C}$$

These rules **internalize weakening**.

Alternative Rules for $\wedge$

We can replace $\wedge R$ with the following rule:

$$\frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B} \qquad \longrightarrow \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}$$

Outline

From Intuitionistic to Linear Logic

- Main idea: treat $\otimes$ the same as $\wedge$, minus contraction and weakening rules.
- The context Γ then externalizes $\otimes$ on the meta-level:

$$\Gamma \vdash A \qquad \longrightarrow \qquad \bigotimes \Gamma \implies A$$

- We will recover contraction and weakening later with the **! modality**.

Proof Theory of $\otimes$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \otimes\mathrm{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B} \qquad \otimes\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\cancel{\mathrm{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C}} \qquad \cancel{\mathrm{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}} \qquad \mathrm{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\otimes$

Left and right rules for $\otimes$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \otimes\mathrm{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B} \qquad \otimes\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\cancel{\mathrm{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C}} \qquad \cancel{\mathrm{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}} \qquad \mathrm{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

Proof Theory of $\otimes$

Sequent calculus: $\Gamma \vdash A$

$$\frac{}{A \vdash A} \qquad \otimes\mathsf{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B} \qquad \otimes\mathsf{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\xcancel{\mathsf{Weakn}\ \frac{\Gamma \vdash C}{\Gamma, A \vdash C}} \qquad \xcancel{\mathsf{Contr}\ \frac{\Gamma, A, A \vdash C}{\Gamma, A \vdash C}} \qquad \mathsf{eXchg}\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}$$

No weakening or contraction

Adding the Unit

A proof theory with just $\otimes$ is not fun.
We need multiplicative unit (1) to signify **absence of resources**:

$$\frac{}{\emptyset \vdash 1} \qquad\qquad \frac{\Gamma \vdash C}{\Gamma, 1 \vdash C}$$

Adding Implication

A proof theory with just $\otimes, 1$ is not fun.
We need linear implication ($\multimap$, *lollipop*) to form linear hypotheticals:

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \qquad\qquad \frac{}{\Gamma_1, \Gamma_2, A \multimap B \vdash C}$$

Adding Implication

A proof theory with just $\otimes, 1$ is not fun.
We need linear implication ($\multimap$, *lollipop*) to form linear hypotheticals:

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \qquad\qquad \frac{\Gamma_1 \vdash A \qquad \Gamma_2, B \vdash C}{\Gamma_1, \Gamma_2, A \multimap B \vdash C}$$

Exercise: try to derive $(A \otimes B) \multimap C \dashv\vdash A \multimap (B \multimap C)$

Cut rule

$$\frac{\Gamma \vdash A \qquad \Gamma', A \vdash B}{\Gamma, \Gamma' \vdash B}\ \text{Cut}$$

What is so special about this rule?

Each cut rule introduces a piece of “complexity” as a new formula A

Cut rule

$$\frac{\Gamma \vdash A \qquad \Gamma', A \vdash B}{\Gamma, \Gamma' \vdash B}\ \text{Cut}$$

What is so special about this rule?
Each cut rule introduces a piece of "complexity" as a new formula A

Theorem

Every proof of $\Gamma \vdash A$ that uses cut can be converted into a proof that does not use cut

MILL

$$A, B \quad ::= \quad 1 \mid A \otimes B \mid A \multimap B$$

$$\frac{}{A \vdash A} \qquad \frac{}{\emptyset \vdash 1} \qquad \otimes\mathsf{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B} \qquad \otimes\mathsf{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\mathsf{Cut}\ \frac{\Gamma \vdash A \qquad \Gamma', A \vdash B}{\Gamma, \Gamma' \vdash B} \qquad \multimap\mathsf{R}\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \qquad \multimap\mathsf{L}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2, B \vdash C}{\Gamma_1, \Gamma_2, A \multimap B \vdash C}$$

Adding &

Recall the different versions of left/right rules for $\wedge$

$$\wedge\mathrm{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \wedge B} \qquad \wedge\mathrm{R'}\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}$$

$$\wedge\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C} \qquad \wedge\mathrm{L}_i\ \frac{\Gamma, A_i \vdash C}{\Gamma, A_1 \wedge A_2 \vdash C}$$

Adding &

Recall the different versions of left/right rules for $\wedge$

$$\otimes\text{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B}$$

$$\wedge\text{R'}\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}$$

$$\otimes\text{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\wedge\text{L}_i\ \frac{\Gamma, A_i \vdash C}{\Gamma, A_1 \wedge A_2 \vdash C}$$

Adding &

Recall the different versions of left/right rules for $\wedge$

$$\otimes\mathsf{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B}$$

$$\&\mathsf{R}\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B}$$

$$\otimes\mathsf{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\&\mathsf{L}_i\ \frac{\Gamma, A_i \vdash C}{\Gamma, A_1 \,\&\, A_2 \vdash C}$$

Interplay of $\otimes$ and &

- $A \mathbin{\&} B \vdash A$, and $A \mathbin{\&} B \vdash B$
- $A \mathbin{\&} B \not\dashv\vdash A \otimes B$
- $(A \multimap B) \mathbin{\&} (A \multimap C) \vdash A \multimap B \mathbin{\&} C$
- $(A \mathbin{\&} B) \multimap C \nvdash A \multimap (B \multimap C)$

Outline

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$E^{15} \multimap \left(\qquad \right)$$

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$E^{15} \multimap \left((Sal \mathbin{\&} Soup) \otimes \qquad \right)$$

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$E^{15} \multimap \left(\begin{array}{l} (\mathit{Sal} \mathbin{\&} \mathit{Soup}) \otimes \\ \big((\mathit{Schn} \mathbin{\&} \mathit{Tofu} \mathbin{\&} (E \otimes E \multimap \mathit{Fish})) \\ \\ \\ \\ \\ \end{array} \right)$$

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$E^{15} \multimap \left(\begin{array}{l} (Sal \mathbin{\&} Soup) \otimes \\ \Big((Schn \mathbin{\&} Tofu \mathbin{\&} (E \otimes E \multimap Fish)) \\ \qquad \otimes \; Fries \Big) \otimes \\ \\ \\ \\ \end{array} \right)$$

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$
E^{15} \multimap \left(\begin{array}{l} (Sal \,\&\, Soup) \otimes \\ \Big((Schn \,\&\, Tofu \,\&\, (E \otimes E \multimap Fish)) \\ \qquad \otimes \;\; Fries\Big) \otimes \\ (Icecr \,\&\, Cheese) \otimes \\ \\ \\ \end{array} \right)
$$

Menu at the Linear Logic Cafe

Summer Menu! €15 p.p.
Starter: Greek Salad, or Soup
Main: Chicken Schnitzel, Tofu, or Salmon (for additional €2)
(all main dishes come with a side of fries)
Desert: Ice Cream, or Cheese Platter
Drinks: Leffe Tripel (for additional €3)

$$E^{15} \multimap \left(\begin{array}{l} (Sal \mathbin{\&} Soup) \otimes \\ \Big((Schn \mathbin{\&} Tofu \mathbin{\&} (E \otimes E \multimap Fish)) \\ \qquad \otimes \;\; Fries\Big) \otimes \\ (Icecr \mathbin{\&} Cheese) \otimes \\ \\ (E^3 \multimap Beer) \end{array} \right)$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\left(\vphantom{\begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \end{array}} \hspace{6em} \right) \multimap Pizza$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\left(\begin{array}{l} \mathit{Yeast} \otimes \mathit{Flour} \otimes \\ \\ \\ \\ \\ \end{array} \right) \multimap \mathit{Pizza}$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\begin{pmatrix} \mathit{Yeast} \otimes \mathit{Flour} \otimes \\ \\ \mathit{Salt} \otimes \mathit{Sugar} \otimes \\ \\ \\ \\ \end{pmatrix} \multimap \mathit{Pizza}$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\left(\begin{array}{l} \mathit{Yeast} \otimes \mathit{Flour} \otimes \\ \\ \mathit{Salt} \otimes \mathit{Sugar} \otimes \\ \\ \mathit{Tomatoes} \otimes \\ \\ \end{array}\right) \multimap \mathit{Pizza}$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\begin{pmatrix} Yeast \otimes Flour \otimes \\ Salt \otimes Sugar \otimes \\ Tomatoes \otimes \\ (Sausage \text{ or } Paprika) \end{pmatrix} \multimap Pizza$$

Grandma's Linear Logic Pizza recipe

LL Pizza Ingredients:

- Yeast and flour,
- Salt and sugar
- Tomatoes
- Sausage or paprika

$$\left(\begin{array}{l} \mathit{Yeast} \otimes \mathit{Flour} \otimes \\ \mathit{Salt} \otimes \mathit{Sugar} \otimes \\ \mathit{Tomatoes} \otimes \\ (\mathit{Sausage} \oplus \mathit{Paprika}) \end{array}\right) \multimap \mathit{Pizza}$$

Adding $\oplus$

$$\oplus\mathsf{R}_1 \quad \frac{\Gamma \vdash A_1}{\Gamma \vdash A_1 \oplus A_2}$$

$$\oplus\mathsf{R}_2 \quad \frac{\Gamma \vdash A_2}{\Gamma \vdash A_1 \oplus A_2}$$

$$\oplus\mathsf{L} \quad \frac{\Gamma, A_1 \vdash C \qquad \Gamma, A_2 \vdash C}{\Gamma, A_1 \oplus A_2 \vdash C}$$

$\oplus$ vs $\&$

The rules are the same as for $\vee$ in the intuitionistic sequent calculus.

$\oplus$ vs $\&$

The rules are the same as for $\vee$ in the intuitionistic sequent calculus.

However,

$$E \,\&\, (B \oplus C) \nvdash (E \,\&\, B) \oplus (E \,\&\, C)$$

- E = one euro;
- B = beer, C = coffee;
- $(B \oplus C)$ = either beer or coffee, but do not know which;
- $E \,\&\, (B \oplus C)$ = I can either have an euro or a drink.

$\oplus$ vs $\&$

$$tea \oplus coffee \vdash awake$$

$$tea \,\&\, coffee \vdash awake$$

$\oplus$ vs $\&$

$tea \oplus coffee \vdash awake$

Given either tea or coffee (I don't care which one), I can drink it and get awake

$tea \,\&\, coffee \vdash awake$

$\oplus$ vs $\&$

$tea \oplus coffee \vdash awake$

Given either tea or coffee (I don't care which one), I can drink it and get awake

$tea \,\&\, coffee \vdash awake$

Given a choice of tea and coffee (for example at a hotel breakfast), I can drink a hot beverage and get awake

$\oplus$ vs $\&$

$tea \oplus coffee \vdash awake$

Given either tea or coffee (I don't care which one), I can drink it and get awake

$tea \,\&\, coffee \vdash awake$

Given a choice of tea and coffee (for example at a hotel breakfast), I can drink a hot beverage and get awake

$tea \,\&\, coffee \vdash tea \oplus coffee$

$\oplus$ vs $\&$

$tea \oplus coffee \vdash awake$

Given either tea or coffee (I don't care which one), I can drink it and get awake

$tea \,\&\, coffee \vdash awake$

Given a choice of tea and coffee (for example at a hotel breakfast), I can drink a hot beverage and get awake

$tea \,\&\, coffee \vdash tea \oplus coffee$

$A \,\&\, B \vdash A \oplus B \qquad\qquad A \oplus B \nvdash A \,\&\, B$

MAILL

$$A, B \quad ::= \quad 1 \mid A \otimes B \mid A \multimap B \mid A \mathbin{\&} B \mid A \oplus B$$

$$\frac{}{A \vdash A} \qquad \frac{}{\emptyset \vdash 1} \qquad \otimes\mathrm{R}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2 \vdash B}{\Gamma_1, \Gamma_2 \vdash A \otimes B} \qquad \otimes\mathrm{L}\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C}$$

$$\mathrm{Cut}\ \frac{\Gamma \vdash A \qquad \Gamma', A \vdash B}{\Gamma, \Gamma' \vdash B} \qquad \multimap\mathrm{R}\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \qquad \multimap\mathrm{L}\ \frac{\Gamma_1 \vdash A \qquad \Gamma_2, B \vdash C}{\Gamma_1, \Gamma_2, A \multimap B \vdash C} \qquad \&\mathrm{R}\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \mathbin{\&} B}$$

$$\&\mathrm{L}_i\ \frac{\Gamma, A_i \vdash C}{\Gamma, A_1 \mathbin{\&} A_2 \vdash C} \qquad \oplus\mathrm{R}_i\ \frac{\Gamma \vdash A_i}{\Gamma \vdash A_1 \oplus A_2} \qquad \oplus\mathrm{L}\ \frac{\Gamma, A_1 \vdash C \qquad \Gamma, A_2 \vdash C}{\Gamma, A_1 \oplus A_2 \vdash C}$$

Related Logics

- Classical Linear Logic

Related Logics

- Classical Linear Logic
- Relevance logics
  - Contraction but no weakening
  - Intuition: all hypothesis are relevant and must be used

Related Logics

- Classical Linear Logic
- Relevance logics
  - Contraction but no weakening
  - Intuition: all hypothesis are relevant and must be used
- Logic of Bunched Implications
  - Better known as the kernel of Separation Logic
  - Popular in deductive program verification

Taking Stock

Up to here:

- Correctness for communicating programs
- Sessions as protocol specifications
- A formal syntax for session types
- Example: A two-buyer protocol
- Protocol compatibility in terms of duality for session types
- **Intuitionistic Linear Logic (MAILL)**

Tomorrow:

- Interpreting ILL as session types for message-passing processes

Cut Elimination as Communication

Asynchronous Communication

Unrestricted Behavior: Clients and Servers

Bunched Implications